Shared geometry for standalone sign windows

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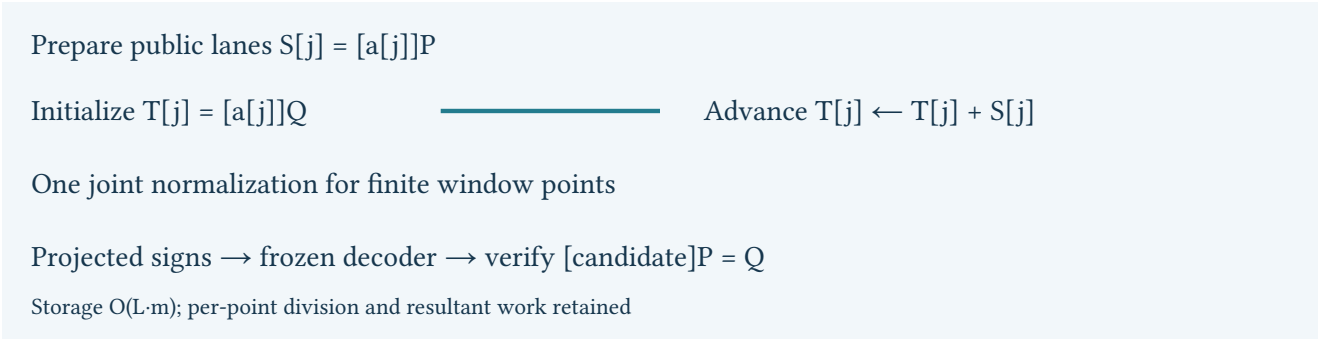
The standalone character source now accepts a jointly normalized window of public point shifts. Recovery uses this route for its initial sign window and retains its final scalar-to-point verification. Conditional extra signs remain standalone queries.



Construction

Prepared public strides and target lanes obey group distributivity at each offset. The borrowed prepared object is tied to the original source and base. Target torsion is checked before geometry construction; the unchanged projected residual and division-factor evaluator rechecks each finite shift. Identity slots contribute sign zero. Window and flattened-point caps precede allocation.

Software evidence

Validation-03 passes four selected native methods; 95 imported methods are filtered out. Forty windows supply 640 sign checks across prime-field fixtures with subgroup orders 7, 13 and 479, including identity crossings. Signs equal both the single-point source and checker scalar characters. All 20 frozen recoveries and 16 extra-sign decisions match, including rejection of changed candidates and a deliberately corrupted decoder.

The normal metadata build excludes six legacy source/transport paths. Thirty-one source/input hashes are pinned. Test and normal builds retain 17 and 23 warnings. Validation-01 preserves the initial included-doc-comment and cursor-type errors; validation-02 preserves the passing window-only stage and exact source snapshot. No independent replay or comparative timing panel was performed.

Cost and limits

A finite query window uses one joint geometry normalization; preparation and calibration have their own costs. Public stride preparation is currently per recovery call. Per-point division caches and resultants remain. Window storage grows as O(L·m), where L is length and m is lane count; the old sequential geometry used O(m). Element-count limits do not claim byte/RSS limits. Prime, native word-size and projection-order restrictions remain unchanged. The original 4,563-case cohort was not rerun.

Next construction and references

Retain prepared strides across recovery calls and continue the final lanes for extra sign offsets. Public compiler enumeration and per-point resultant work remain the larger scaling tasks. DESIGN.md and validation-03/receipt.json provide the mechanism and replay evidence. Sources: existing joint-window, standalone-source and recovery modules; official Rust lifetime documentation at https://doc.rust-lang.org/book/ch10-03-lifetime-syntax.html. KB unavailable.